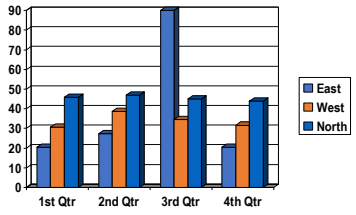
Lesson 8

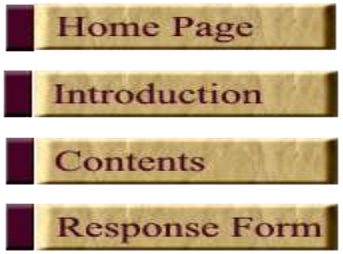
Graphics file formats

Graphics

- **Graphics** - anything on the page that is not actual text, from simple line drawings to fully active images found on the World Wide Web.
- There are many different graphic file formats.
- The format you choose depends on several factors.



Chart



Buttons

Photograph

Factors that Affect Graphic Formats

- Type of graphics
- Colour Depth
- Compression
- Portability
- Transparency

Types of Graphics

- There are two basic types of graphics:
  - *Bitmapped* and
  - *Vector*

- Bitmapped graphics are much more common
- Often they are called raster graphics
- When you create a bitmapped graphic you are basically creating a bunch of coloured dots

Bitmapped Graphics

- The bitmapped graphic is stored as an array / grid of dots, or pixels
- Each pixel gets assigned a specific colour
- The more pixels, the more detailed the image can be
- Can be a continuous-tone image, such as a photograph
- Bitmap graphic editors are called paint programs
- Enlarging a bitmap graphic may cause the image to lose crispness and clarity
- Some common bitmap graphics programs are:
  - Photoshop
  - Paint Shop Pro
  - Photo-Paint
  - Graphic Converter

Bitmap Images

- Very flexible
  - Any image can be represented (with enough pixels)
- Created by scanners, digital cameras, and other similar devices
- Used most commonly, especially on the web
- Can be displayed directly on your computer screen if 1 image pixel is the same size as 1 screen pixel
- Takes a lot of memory
  - The colour of each pixel must be stored
  - Can be compressed

Vector Graphics

- It describes the image using shapes i.e. Circles, Lines and curves
- Also has to store the colour of these shapes
- Generally much smaller than raster graphic files
- Vector graphic editors are called drawing programs e.g.:
  - Corel Draw
  - Adobe Illustrator
  - Acrobat
- The images do not lose clarity as you enlarge them.
- Appear as bitmaps on computer monitors because computer monitors consist of pixels

Vector Images

- Easy to change parts of the image since each part is stored as a different shape
- Can manipulate the image smoothly
  - Rotating, changing colour, size, line width, etc.
- Limited to the shapes the program can handle
- Typically takes less memory and disk space than a bitmap
- Must be converted to a bitmap to display

Colour Depth

- The number of colours per pixel that can be contained in an image.
- Most graphics editing programs will allow you to set the colour depth for your image.
- Different graphic formats contain different numbers of colours per pixel.
- Examples:

1-bit colour	2 colours Per Pixel
8-bit colour	256 colours Per Pixel
16-bit colour	65.5 Thousand colours Per Pixel
24-bit colour	16.7 Million colours Per Pixel

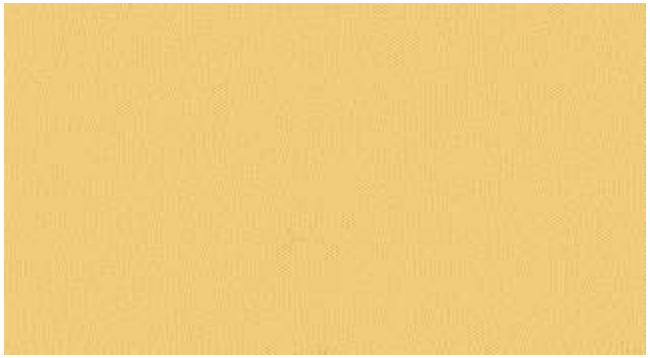
Colours vs. Image Size

- It’s nice to use as many colours as possible, especially when dealing with photographs
- But, the more colours, the larger the image is to store
  - More information is required for each pixel
- Can work in 24-bit colour then save the image with fewer colours for storage.
- What if your image has more colours than the colour depth?
  - If the colour is not in the palette, the program can choose a separate, close colour, or do dithering
  - Dithering is where the program fools your eye by creating a pattern of pixels in two colours which makes you see a colour between the two

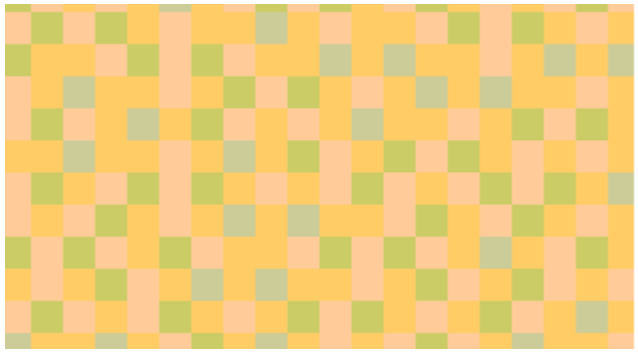
Dithering with colour

colour I want to produce

Dithering with only 2 colours, orange and white



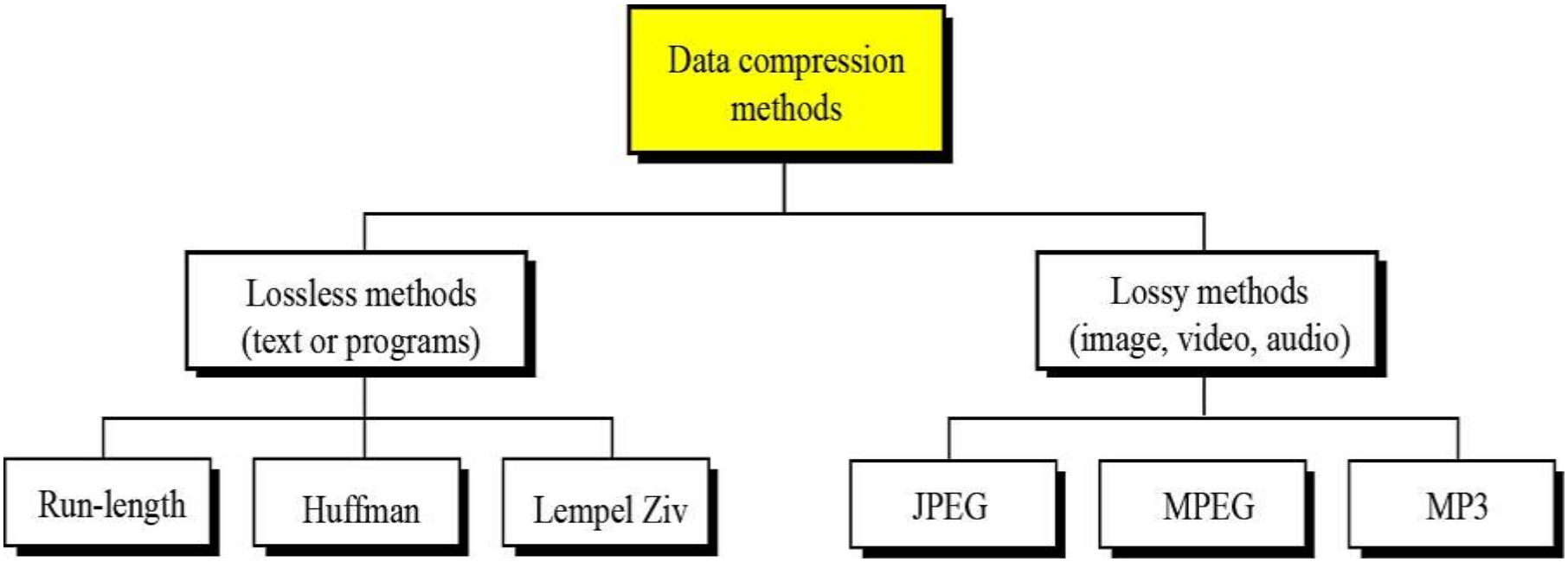
Dithering with all web-safe colours enabled



Zoomed View of dithering

Data Compression Methods

- Data compression is about storing and sending a smaller number of bits.
- There're two major methods used to compress data: lossless and lossy methods



Lossy vs. Lossless Compression

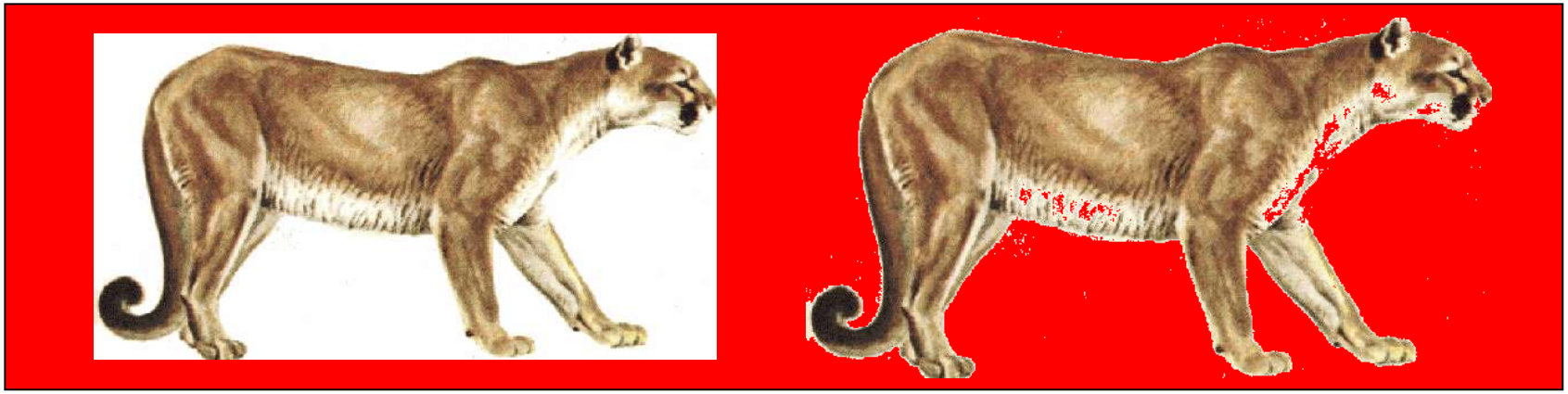
- When an image is compressed and later uncompressed with a *lossless algorithm*, the image is exactly the same
    - Reduces the file size without losing any pixel data
    - Quality is not compromised
- *Lossy algorithms* produce images which may be slightly different before compression and after uncompressing
    - Some information may be lost
    - Can sometimes produce much smaller files than a lossless algorithm
    - Deletes some of the data that it deems unnecessary since the human eye is not very sensitive to changes in colour over small distances.
    - The more the image is reduced, the more the quality is degraded

Portability

How easily you can open, modify, and view the files on computers using different operating systems, software and browsers.

Transparency

- Allows the background colour of an image to be “eliminated” or made transparent so that the background behind the image can be seen.
- It makes part of the image invisible, or “see-through.”
- Often used with graphics on web pages
  - The image is still a rectangle. Parts of the background show through so it doesn’t look like a rectangle



Simple Transparency

- Some pixels are marked as transparent
  - The background will show through
- Images can appear any shape
- It’s “all or nothing” transparency
  - Either the pixel is completely transparent or its not

Alpha Channel

- *Alpha channel* is a general method of transparency
- Images have two parts:
  - The image
  - A mask, called the alpha channel
- The alpha channel indicates how transparent each pixel is
- If supported by the program and format, you can specify partial transparency of images
- Photoshop supports full transparency
- So does the PNG format
- If it’s not supported, the advanced transparency information is ignored

File Formats

- After picking a graphics format we need to select a file format to use in storing images.
- Each way of storing an image is called a file format
- Each file format converts the image to a corresponding string of bits differently in order to store them on disk or transmit them over the Internet

How to select a file formats

- **Type**: does the file format store the type of image you want to use (bitmap or vector)?
- **Colour Depth**: does the format support the number of colours you need?
- **Compression**: can make the file smaller, but it takes time to compress and decompress a file
- **Portability**: can other people use images in this format?
- **Transparency**: do parts of your image need to be transparent?

Common File Formats

Bitmap Formats

- GIF: graphics interchange format
- JPEG: joint photographic experts group
- PNG: portable network graphic
- BMP: Windows bitmap
- TIFF: tagged image file format

Vector Formats

- SVG : scalable vector graphics
- EPS: encapsulated postscript
- CMX: Corel meta exchange
- PICT: Macintosh Picture
- WMF: Windows metafile

JPEG

- JPEG : Joint Photographic Experts Group
- .jpg is the file extension (normally)
- It was designed for photographs at high quality and in full colour
- Lossy compression format
- Bitmap format
- Normally changes can’t be seen except at the lowest quality settings
- Bad idea to store simple black and white files as JPEGs
- Good for 8-bit grayscale or 24-bit colour
  - Like you’d get from a digital camera or scanner

GIF

- GIF : Graphics Interchange Format
- .gif file extension
- Bitmap format
- Lossless compression
- Used a lot on the web
- Good for simple animations and simple transparency
- Limited to 8-bit colour (256 colours)
- Compression algorithm is patented

PNG

- PNG : Portable Network Graphic
- Bitmap format
- Lossless compression
- Replacement for GIF (no patent, thus FREE)
- Better compression algorithm than GIF
- Supports 24-bit colour and full transparency